
Artificial Intelligence and Academic Integrity: Legislate or Educate?

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This participatory action research intends to bridge the gap between generative artificial intelligence (GenAI) tools and academic integrity in a Midwestern university in the United States. Using mixed methods, researchers analysed participants' perceptions and strategies for using GenAI tools while upholding academic integrity. The study highlighted the institution's strengths, weaknesses, opportunities, and challenges in grappling with the issue. The results pointed to the need for the critical use of GenAI tools and guidelines for tackling the multifaceted impacts of technology. The study contributes to the field with strategies to empower educators and learners in navigating the ethical landscape of GenAI in education.

Keywords: artificial intelligence, academic integrity, critical artificial intelligence literacy, generative artificial intelligence, participatory action research

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INTRODUCTION

Artificial intelligence (AI) refers to systems designed to *simulate* human intellectual capabilities, such as reasoning, problem-solving, and decision-making. AI has unprecedentedly changed how people live, teach, learn, and research. However, effective and ethical AI usage requires users' critical awareness of the nature of AI systems, which are, in fact, information processors rather than original creators. Generative AI (GenAI) is a type of AI trained on massive amount of data. It can generate content in various formats, including text, images, music, and other media, by processing information through advanced techniques like neural networks and machine learning.

The release of ChatGPT by OpenAI in November 2022 marked the start of a significant era of general access to GenAI technology (Perkins, 2023). Since then, the proliferation of various GenAI tools has fuelled powerful applications across diverse domains, including work, research, and learning experiences (Currie, 2023; Moorhouse et al., 2023). Emerging evidence suggests that integrating GenAI tools into the learning process leads to a significant shift in educational practices, affecting multiple realms, including academic integrity (Bahroun et al., 2023; Cotton et al., 2023; Perkins, 2023; Perkins & Roe, 2023; Rahman & Watanobe, 2023; Rudolph et al., 2023; Uzun, 2023). Academic integrity is a commitment to 'six fundamental values: honesty, trust, fairness, respect, responsibility, and courage' (International Centre for Academic Integrity, 2021, p. 3) in which every member of an academic community—students, staff, faculty members, and administrators—is accountable for safeguarding the integrity of its scholarship, teaching, research, and service. However, the overwhelming number of GenAI tools has outpaced adapting and regulating ethical and effective frameworks (UNESCO, 2023).

Although GenAI tools may produce content with impressive writing mechanics, GenAI outputs may include significant inaccuracies that could go unnoticed if not thoroughly checked. Inaccuracies may occur because of the training data and algorithmic biases that reflect inequalities and biases inherent in human writing (Jandrić, 2019). Consequently, GenAI technology has brought about daunting challenges for higher education institutions (HEIs) in their pursuit of upholding academic standards and equitable assessments.

This research investigates stakeholders' perceptions and uses of GenAI tools and their impact on academic integrity at one urban Midwestern university in the United States. This participatory action research (PAR) examines academic integrity in the age of AI, acknowledging the multifaceted impact of GenAI tools. Taking a PAR approach, the researchers collect data from multiple university stakeholders through surveys, focus groups, and the PAR team analysis process. This project seeks to answer these questions:

- RQ1: What are stakeholders' perceptions and knowledge of GenAI tools in teaching and learning?
- RQ2: What are stakeholders' perceptions of academic integrity in the AI era?
- RQ3: How effective are the existing university guidelines, policies, practices, and strategies for academic integrity and GenAI usage?
- RQ4: What actions are needed to prepare the institution for the age of AI?

This PAR's findings inform stakeholders with recommendations on upholding academic integrity while integrating these emerging technologies to transform teaching and learning. By presenting a prompt empirical study and an evolving framework for critical AI literacy (CAIL) education, this research contributes to the expanding intersection of AI and academic integrity.

LITERATURE REVIEW AND THEORETICAL FRAMEWORK

AI: To Educate or Legislate?

AI research has been an ongoing field in computer science since discussions on human-like machine intelligence appeared in the late 1940s (McCarthy, 2007). However, more recently, the advancement of GenAI technology has fuelled some controversial debates in education. While some stakeholders ban GenAI tools altogether, others embrace the opportunities they present (Bahroun et al., 2023; UNESCO, 2023). Considering different approaches to GenAI technology, this study explores whether upholding academic integrity through critical AI training (*educating*) or regulating AI usage (*legislating*) within academia is more effective.

Research has addressed how institutional policies can support ethical AI conduct. According to UNESCO's (2023) policy guidelines and framework,

approaches to GenAI should be human-centred, where human capabilities are enhanced by technology, not constrained nor replaced. Additionally, Chan's (2023) AI Ecological Education Policy Framework acknowledges the crucial role of stakeholders such as students, faculty, staff, management, and external agents to ensure responsibility and awareness. The framework (Chan, 2023) is organised into three dimensions: 1) Pedagogical (enhancing teaching and learning goals through responsible use of AI); 2) Governance (dealing with issues of AI access, privacy, security, and accountability); and 3) Operational (providing and assessing AI implementation and training). Thus, both UNESCO (2023) and Chan (2023) are aligned because they require a humanistic, intersectoral, and interdisciplinary approach to comprehensive and ethical AI implementation.

To examine how Australian public universities adapted to the rise of GenAI tools, De Maio (2024) updated an existing academic integrity framework (Bretag et al., 2011) to address the challenges and opportunities presented by GenAI technology. The original framework consisted of well-defined principles for academic integrity, that is: 1) access, 2) approach, 3) responsibilities, 4) detail, and 5) support). With the rapid development and popularity of GenAI, De Maio (2024) recognised the need to acknowledge advancements, demands, and concerns posed by GenAI and added two new categories: 6) currency and 7) flexibility. Thus, this updated framework emphasises the importance of regular policy updates and AI education for faculty, staff, and students to ensure ethical GenAI usage. As more scholars join the conversation and seek to mitigate GenAI's potential threats to academic integrity (Birks & Clare, 2023; Cotton et al., 2023; Teel et al., 2023), there is a general agreement that university stakeholders need to learn how to use GenAI tools ethically and responsibly proactively. This challenge necessitates the development and public access to critical education about AI.

CAIL: An Emerging Framework

Integrating AI literacy into K–12 and university education is a recent development, surfacing only within the past few years (Long & Magerko, 2020). Research has built the concept of AI literacy expanding from the wider literacy umbrella (Ng et al., 2021), acknowledging four aspects to frame it: 1) Knowing and Understanding; 2) Using and Applying; 3) Evaluating and Creating; and 4) Ethical Issues. While GenAI offers exciting

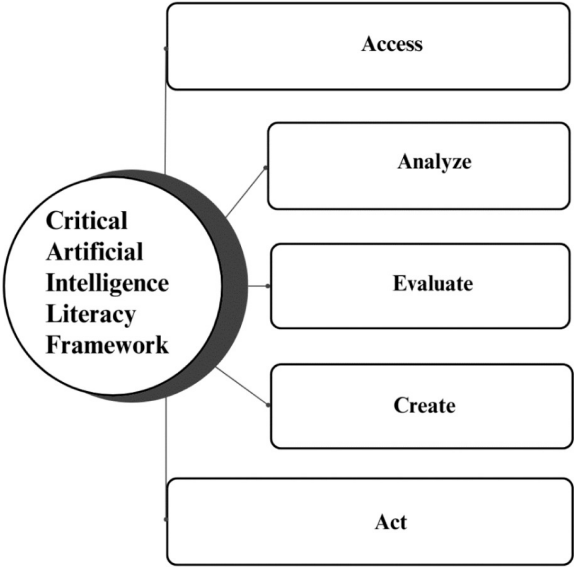


FIGURE 1. Critical Artificial Intelligence Literacy Framework

possibilities for enriching educational practices, concerns persist regarding biases, misuse, and its impact on student learning.

In response, the authors have developed a Critical Artificial Intelligence Literacy (CAIL) framework to address the challenges of GenAI technology and support its ethical integration into education and research. This framework integrates insights from existing models in AI literacy (Long & Magerko, 2020; Ng et al., 2021), AI policy implementation (Chan, 2023; UNESCO, 2023), and critical media literacy (CML; Hobbs, 2010). CML, which emphasises analysing and evaluating media, provides a foundation for CAIL, allowing the integration of GenAI tools in education while upholding academic integrity. This emerging framework requires the development of basic skills for navigating GenAI ethically. In Figure 1, these skills refer to

1. basic skills to access GenAI: what GenAI tools are out there, where to find them, who can use them, for what purposes, and how to best use GenAI applications effectively and responsibly.
2. basic skills to analyse GenAI tools and their outputs: identifying their developers, purposes, and inherent biases. Hence, AI literacy

encompasses awareness of basic inner system functionalities, trained data, privacy concerns, and security issues.

3. basic skills to evaluate the social and political implications of GenAI: recognising AI's broader issues, including environmental impacts, worker exploitation, labour displacement, and data poverty. This awareness helps us understand how AI might influence social structures, political processes, and resource access.
4. basic skills to create new content: applying GenAI tools and technologies effectively, ethically, and critically. Ethical creations with GenAI tools require practices that promote critical thinking skills about AI-generated content.
5. the courage to act through reflection: reflecting on one's conduct and behaviour by applying social responsibility and ethical principles, taking social action by creating a critical AI-literate environment to uphold academic integrity and transform research, learning, and teaching practices.

Drawing on critical pedagogy (Freire, 2000), CAIL can be an approach to ethically integrate GenAI tools into teaching and learning praxis that confronts the educational systems with inherent biases that threaten to subjugate already marginalised people and communities. Because GenAI is reshaping how educators teach and how students learn (Bahroun et al., 2023), educational policies must be constantly adapted to not only address the ethical threats posed by GenAI but also optimise the opportunities for greater equity, human-centred and creative knowledge creation in Education (UNESCO, 2023). The preceding CAIL framework guided the researchers in forming their goals, research questions, data collection, interpretation of results, and recommendations.

METHODOLOGY

Research Design

In exploring whether to educate or legislate for AI tools at a US Midwestern university, this PAR project attempts to bridge the gap between GenAI and academic integrity. Thus, this research plays an active and influential role in decisions that affect participants' lives (Chevalier & Buckles, 2019). Taking a mixed-method approach, the PAR team, alongside other university stakeholders, analysed their perspectives and strategies on GenAI ethical usage. This approach helps 'to understand the holistic

picture from meanings obtained from interviews to the prevalence of traits in a population obtained from surveys, which add depth and breadth to the study' (Wasti et al., 2022, para. 4), and it enriches the understanding of the phenomena. Hence, a convergent mixed-methods approach was adopted with concurrent collection of quantitative data from faculty, staff, and student surveys; qualitative data from faculty, staff, and student focus groups; and collaborative discussions and memos within the PAR team. Hence, the research was designed to discover ways to improve and meet the challenges and needs of upholding academic integrity in the AI era.

Collaboratively, the team identified the critical areas of inquiry: examine stakeholders' perceptions and knowledge of GenAI tools in teaching and learning (RQ1); stakeholders' perceptions of academic integrity in the AI era (RQ2); explore existing university guidelines, policies, practices, and strategies for academic integrity and using GenAI tools (RQ3); and propose potential actions that may prepare the institution for the age of AI (RQ4).

Participants

This PAR project involved the research team (i.e., three faculty members, one administrator, one librarian, and two doctoral students), survey respondents, and focus group participants. To ensure fruitful discussions, the research team employed purposive sampling to recruit participants who were seriously impacted by and concerned about GenAI for the focus groups. This technique allowed us to strategically involve university stakeholders with diverse demographics, knowledge, and backgrounds who could best contribute to the study (Creswell & Poth, 2018). In addition, snowball sampling (Patton, 2015) allowed the team to reach a wider audience. Through a mix of purposeful and snowball recruitment processes, the stakeholder rainbow analysis (Chevalier & Buckles, 2019) indicated diverse participants involved in the focus groups, such as twelve faculty, staff, and administrators; four undergraduate students; and five graduate students (see Figure A1 in the Appendix). Hence, there was a balance in the participants' selection considering their influence as university stakeholders. Seven focus groups with twenty-one university stakeholders conducted over two days yielded rich information.

Multiple rounds of recruitment were needed to achieve the minimum sample numbers for the surveys. Specific actions to recruit survey respondents included but were not limited to the announcement at an online

webinar focused on AI with follow-up to all registrants, inclusion in the teaching and learning centre's newsletter (to 2400+ electronic mailing service), targeted email invitations to the university's academic integrity task force and other faculty and administrator colleagues across campus, and in graduate and undergraduate student newsletters, all of which included the invitation for individuals to spread the call to their networks. These recruitment efforts yielded sixty-two faculty and staff and twenty-seven student participants for the surveys. While survey participation fell short of initial expectations, the focus group discussions provided robust and insightful information. Qualitative data from focus groups helped capture a detailed description of the issue under study, and the analysis process allowed researchers to draw conclusions (Manfra, 2021).

Instruments and Data Analysis

This project utilised multiple instruments to collect data on GenAI practices and perceptions. Data instruments included PAR team discussion notes, focus group recordings, and online survey responses. A critical thematic analysis was then conducted to identify key themes within this rich data set. Details of the research approach are described in the following sections.

Quantitative Instruments and Data Analysis

Two online survey questionnaires with the same items were created (i.e., one for faculty and staff and another for students) to collect stakeholder perceptions and practices on AI. Each questionnaire had thirty-one items: twenty-three 6-point Likert-scale questions, two open-ended questions, and six multiple-option demographic questions (see Table A1 in the Appendix for original survey items). PAR team members piloted the questionnaires and provided feedback. As a result, some wording in the questions and the format were changed for clarity.

After the data collection, descriptive analysis was first performed on numerical data from Questions 1–23 and categorical data from Questions 26–31 to understand data distribution trends and variances among the responses. Then, for all numerical data, Spearman's correlation (Hauke & Kossowski, 2011) analysis was used to find both strong correlations (>0.95 or <-0.95) and very weak or non-existing correlations (between -0.05 and 0.05). In addition, group-based analysis was done using the chi-squared test (Witkov & Zengel, 2019) for independence to find

potentially different behaviours from diverse groups from the categorical data, where the researchers grouped all responses, with an agreeing nature in one group and all responses with a disagreeing nature in another group.

For further inferential analysis, data from the faculty and staff survey were analysed with dimension-reduction techniques, specifically factor analysis, to identify latent structures within the data set. Subsequent hypothesis testing was conducted using the Kruskal–Wallis test (Daniel, 1990), a non-parametric method suitable for comparing independent groups without assuming a normal distribution. This method is particularly appropriate given the data's restrictions to provide insights into the survey's underlying patterns.

Qualitative Instruments and Data Analysis

Qualitative instruments included student, faculty, and staff focus groups, open-ended online survey questions, and the PAR team's discussion notes. Focus group prompts included nine semi-structured, open-ended questions, allowing PAR team members to ask follow-up questions. The researchers reviewed the questions before collecting data from surveys and focus groups to allow for feedback. Focus groups were conducted and recorded in audio and video formats, with auto transcriptions generated upon participants' consent. These recordings provided valuable resources for the researchers and eliminated the need for manual transcription. Various PAR tools, such as brainstorming exercises, stakeholder rainbow, and Strengths, Weaknesses, Opportunities, and Threats (SWOT) analyses (Mind-Tools, n.d.), facilitated collaborative discussions, reflections, and collective decision-making among research participants, which led the PAR team to create a research-based action plan.

A thematic analysis with an open-ended inductive approach to data was adopted for this study. As Merriam (2009) suggests, 'making sense out of data involves consolidating, reducing and interpreting what people have said and what the researcher has seen and read' (pp. 175–176). The analysis involved 1) reading and coding the open-ended questions and focus group transcripts, 2) finding key concepts, ideas, and themes in the data, and 3) defining and codifying essential ideas and themes in the coding sheet. Each data set was coded by one researcher first and then reviewed with at least one other independent coder. The PAR team

reviewed each data set for emerging patterns and relationships among themes (Namey & Trotter, 2015). These themes were used to triangulate and confirm the findings.

To ensure the study's validity and trustworthiness, the researchers trained the PAR team in qualitative data analysis to achieve acceptable inter-rater reliability. Two PAR members cleaned the data sets, created their codes independently, and later compared their reports. They then worked together to define the codes and recode the data for consensus. Using an open-ended and iterative process, researchers coded the data and created themes based on three or more similar codes from all the data sources.

The PAR team reduced the massive data to one page with significant themes related to the research questions, which was reviewed alongside the quantitative data analysis through the PAR process for interpretation and triangulation purposes. The PAR tools used for this research involved the PAR team through discussions and reflections, supporting the interpretations and sensemaking of the qualitative and quantitative findings. Finally, the research team shared preliminary findings with focus group participants and ensured their voices were accurately represented.

RESULTS

This study investigated stakeholder perceptions and knowledge regarding GenAI in education, academic integrity in the context of AI, the effectiveness of current GenAI policies, and necessary actions to improve CAIL. The researchers conducted surveys for quantitative data and focus groups for qualitative depth. The PAR framework facilitated an inclusive, stakeholder-centred exploration of the topic. A comparative analysis was conducted, followed by merging and interpreting the data through the PAR lens.

The following sections present the quantitative and qualitative data analysis results separately. The quantitative results and analysis of the two surveys are discussed in the survey analysis for faculty and staff and survey analysis for students, respectively. The qualitative results from the two open-ended questions on the surveys, insights from focus groups, and discussions within the PAR team are discussed in the qualitative analysis section.

Quantitative Results: Survey Questions

Quantitative analysis was conducted on each survey to examine students', faculty's, and staff's attitudes and perceptions about using GenAI in educational settings. While the sample size limited extensive inferential statistical analysis, it nonetheless served as a valuable exploratory tool for understanding the underlying patterns in the collected data. The descriptive analyses laid the groundwork for assessing the distribution of responses, offering preliminary insights into the general inclinations and variances in viewpoints among the surveyed parties.

Upon further analysis through factor analysis and hypothesis testing via Kruskal–Wallis tests (Daniel, 1990), significant differences emerged, particularly when examining responses about effective AI policies and academic integrity guidelines. These differences were pronounced among various academic units, indicating a range of attitudes from neutral to strong disagreement with the effectiveness of the current policies that dismiss addressing academic integrity in the age of AI. Moreover, when looking at the need for AI institutional support, training, and awareness, there were no significant differences based on gender, age, leadership role, and the highest degree obtained, suggesting a uniformity in perception across these demographic factors.

While showing a different distribution of responses compared to faculty and staff, the student survey data, particularly concerning the effectiveness of current AI guidelines and academic integrity, also indicated a trend toward disagreement or neutrality. Correlations among the student responses were generally weak to non-existent, indicating students' personal AI strategies were not influenced by whether there are institutional guidelines and training for ensuring academic integrity when using AI tools in educational activities.

Analysis of Survey Responses from Faculty and Staff

The reliability analysis of a scale with thirty-eight valid cases is shown in Table A2 (in the Appendix), with a detailed internal consistency analysis for the faculty and staff survey. The scale's reliability was high, with a Cronbach's (1951) alpha of .881, indicating good internal consistency among the items. When standardising the items, Cronbach's alpha slightly increased to .890, which further signified that the scale's items consistently measured the same underlying construct.

Descriptive statistics of the faculty and staff survey (see [Table A3](#) in the [Appendix](#)) showed responses to survey items ranging from ‘1—Strongly Disagree’ to ‘6—Strongly Agree,’ with an extra option ‘0—N/A, or I do not know’ (original survey questions are shown in the [Appendix](#), [Table A1](#)). The descriptive statistics excluded all responses of ‘0.’ Comparatively, responses about current AI guidelines were skewed towards ‘Disagree,’ with scores in the 2s (RQ3). In comparison, responses about the need for updated academic integrity policies were skewed towards ‘Agree,’ with scores in the upper 3s and above 4s (RQ2). Understanding the benefits of AI tools was in the upper 3s, while using strategies to embrace AI was in the low 3s (RQ1). Using strategies for academic integrity was almost 5, while concerns for AI tools were equally high (4.92; RQ2).

Kurtosis is a statistical measure that describes the tails of a distribution relative to a normal distribution, which is symmetrical and has a kurtosis of zero. If the kurtosis deviates noticeably from zero, the distribution is either flatter or more peaked than usual. Considering Bergman et al.’s (2001) confirmatory factor analysis, which used survey data with skewness values ranging from ± 2.5 , the faculty survey data set met the symmetry requirements for parametric analysis (within 2.5), such as factor analysis. Refer to [Table A2](#) (in the [Appendix](#)) for detailed data on the descriptive statistics for the faculty and staff survey.

Factor Analysis: Analysing the Kaiser–Meyer–Olkin (KMO) measure and Bartlett’s test ([Shrestha, 2021](#)) outputs suggested that the data were moderately suitable for factor analysis (see [Table A4](#) in the [Appendix](#)). The KMO measure was 0.660, a middling value, indicating a moderate level of common variance among the variables. Although not strong, it is adequate for factor analysis. In addition, Bartlett’s test of sphericity showed a chi-squared value of 802.420, with 253 degrees of freedom and a significance level below .001, indicating that the variables were sufficiently correlated for factor analysis. Overall, these tests indicated that despite lower sampling adequacy, significant relationships among variables suggest that factor analysis was informative for the data sets.

The output from a principal component analysis (PCA; [O’Rourke & Hatcher, 2013](#)) indicated that each component progressively explained the variance in the data (see [Table A5](#) in the [Appendix](#)). The first five components accounted for approximately 69.4 per cent of the variance, suggesting that they captured most of the information in the data set. These data

suggest that a reduced number of components could be used to simplify the model without losing significant information, which is a common practice in dimensionality reduction to avoid overfitting and enhance the interpretability of the data.

To further interpret the findings from the rotated component matrix with the extraction method of PCA (O'Rourke & Hatcher, 2013), Table 1 shows the components extracted from the data and the survey items with the highest factor loadings on each component. The following themes emerged:

- Component 1: AI Resources and Institutional Support
- Component 2: Clear and Up-to-Date Academic Integrity Guidelines
- Component 3: Training on Academic Integrity in the Age of AI
- Component 4: Academic Integrity and AI Guidelines
- Component 5: Awareness of AI Benefits and Concerns and Academic Integrity Strategies

Kruskal–Wallis Analysis of Factor Components across Independent Variable Categories: The independent-samples Kruskal–Wallis test (Daniel, 1990) is a non-parametric method for testing whether samples originate from the same distribution. It is used when the assumptions of one-way analysis of variance are unmet, such as when the sample sizes are very small or unequal within independent variable sub-categories. The Kruskal–Wallis test includes more than two groups. It ranks all the data from all groups together and then evaluates whether the distribution of ranks is the same across the groups. If the p -value from the Kruskal–Wallis test is less than the chosen significance level, then the null hypothesis that the groups have the same distribution is rejected.

For each of the five-factor components tested, the null hypothesis was that the distribution is the same across gender, age group, leadership role, affiliated unit, and highest degree. The statistical analysis in Table 2 suggests no statistically significant differences in the distribution of the factor scores by these independent variables in the context of the tested dependent variables. However, there were two exceptions: Participants from different affiliated units showed significant differences for Component 2 (i.e., Clear and Up-to-Date Academic Integrity Guidelines) and Component 4 (i.e., Academic Integrity and AI Guidelines).

Findings Based on Factor Analysis—Significant Differences Concerning Affiliated Units and Their Tendencies: The faculty ratings on

TABLE 1. Rotated Component Matrix with the Extraction Method of PCA: Varimax with Kaiser Normalisation Rotation Method

Rotated Component Matrix					
Statements from Survey Items (Appendix Table A1)	Components				
	1	2	3	4	5
Q3. Whether the Department/Unit has provided clear guidelines for using artificial intelligence tools in educational activities.	0.861	0.316	−0.138	0.035	0.103
Q6. Whether the Department/Unit has provided up-to-date guidelines for using artificial intelligence tools in educational activities.	0.846	0.307	−0.059	0.069	0.140
Q5. Whether the College/School has provided up-to-date guidelines for using artificial intelligence tools in educational activities.	0.815	0.195	0.320	0.158	0.042
Q2. Whether the College/School has provided clear guidelines for using artificial intelligence tools in educational activities.	0.790	0.188	0.229	0.028	0.052
Q1. Whether the University has provided clear guidelines for using artificial intelligence tools in educational activities.	0.751	0.064	0.095	0.043	0.143
Q4. Whether the University has provided up-to-date guidelines for using artificial intelligence tools in educational activities.	0.739	0.031	0.287	0.372	0.044
Q9. Whether the Department/Unit offers training for using artificial intelligence tools in educational activities.	0.576	−0.129	0.265	0.336	0.279
Q14. Whether the College/School has up-to-date guidelines for academic integrity in educational activities.	0.140	0.872	0.164	0.249	−0.110
Q15. Whether the Department/Unit has up-to-date guidelines for academic integrity in educational activities.	0.348	0.834	−0.114	0.126	0.003

(Continued)

TABLE 1. Continued

Statements from Survey Items (Appendix Table A1)	Rotated Component Matrix				
	Components				
	1	2	3	4	5
Q13. Whether the University has up-to-date guidelines for academic integrity in educational activities.	0.162	0.800	0.206	0.115	0.113
Q12. Whether the Department/Unit has clear guidelines for academic integrity in educational activities.	0.329	0.587	−0.267	0.525	0.092
Q16. Whether the University offers training regarding academic integrity in educational activities.	−0.061	0.033	0.805	0.164	0.085
Q7. Whether the University offers training for using artificial intelligence tools in educational activities.	0.488	0.125	0.763	−0.077	0.115
Q8. Whether the College/School offers training for using artificial intelligence tools in educational activities.	0.309	−0.012	0.743	−0.274	0.230
Q10. Whether the University has clear guidelines for academic integrity in educational activities.	0.192	0.135	−0.029	0.785	0.210
Q11. Whether the College/School has clear guidelines for academic integrity in educational activities.	0.111	0.506	0.097	0.752	−0.071
Q20. Whether the respondent understands the concerns about using artificial intelligence tools in educational activities.	0.175	−0.012	0.096	0.283	0.813
Q23. Whether the respondent uses strategies to ensure academic integrity in educational activities.	−0.044	0.308	0.109	−0.027	0.806
Q19. Whether the respondent understands the benefits of using artificial intelligence tools in educational activities.	0.358	−0.319	0.178	0.024	0.680

(Continued)

TABLE 1. Continued

Statements from Survey Items (Appendix Table A1)	Rotated Component Matrix				
	Components				
	1	2	3	4	5
Q22. Whether the respondent uses strategies to embrace artificial intelligence tools in educational activities.	0.316	-0.168	0.365	-0.412	0.460
Q18. Whether the Department/Unit offers training regarding academic integrity in educational activities.	0.189	0.272	0.000	-0.025	0.115
Q17. Whether the College/School offers training about academic integrity in educational activities.	-0.017	0.251	0.532	0.323	-0.022
Q21. Whether the respondent supports restricting the use of artificial intelligence tools in educational activities.	-0.140	0.258	-0.061	-0.081	0.005

Note: Rotation converged in eleven iterations. Data in bold indicates factor loading values of major contributing items for Components.

Component 2 (i.e., Clear and Up-to-Data Academic Integrity Guidelines) displayed significant variation across different academic units. Specifically, the arts and sciences reported a mean (*M*) rating of 4.00, with a standard deviation (*SD*) of 1.109, showing mild agreement with current academic integrity guidelines in the age of AI. These results contrasted with the Affiliated Unit Not Listed, which had a much lower mean rating of 1.91 and a higher standard deviation of 1.921, suggesting a tendency towards strong disagreement. The nursing faculty's ratings were moderately in agreement, with a mean of 3.50 and a standard deviation of 2.121. At the same time, the School of Education faculty had a mean of 3.00 and a standard deviation of 2.380, showing a neutral stance towards academic integrity guidelines. Finally, the continuing and professional studies faculty showed strong disagreement, with the lowest mean rating of 0.50 and a standard deviation of 0.707. On a scale where 6.00 denoted strong agreement and 1.00 denoted strong disagreement, the overall indication is that faculty members' sentiments ranged from neutral to solid

TABLE 2. Kruskal–Wallis Test Assessing Gender Differences in Factor Scores

Null hypothesis	Gender sig. ^{a,b}	Age group sig. ^{a,b}	Leadership role sig. ^{a,b}	Affiliated unit sig. ^{a,b}	Highest degree sig. ^{a,b}
1. AI Resources and Institutional Support	.064	.370	.402	.471	.191
2. Clear and Up-to-Date Academic Integrity Guidelines	.266	.214	.583	.017**	.195
3. Training on Academic Integrity in the Age of AI	.583	.663	.284	.437	.838
4. Academic Integrity and AI Guidelines	.227	.611	1.000	.038**	.306
5. Awareness of AI Benefits and Concerns and Academic Integrity Strategies	.436	.896	.212	.571	.211

Note: Data in bold indicate statistically significant differences in factor scores.

^a The significance level is .050.

^b Asymptotic significance is displayed.

** Significant difference.

disagreement concerning having clear and up-to-date academic integrity guidelines (RQ 3).

The analysis of Component 4 (i.e., Academic Integrity and AI Guidelines) also revealed significant variations across different academic units regarding academic integrity guidelines in the age of AI. The arts and sciences faculty rated it with a mean of 4.00 and a standard deviation of 1.797, showing general agreement with some variability. The Affiliated Unit Not Listed reported lower agreement with a mean score of 2.66 and a standard deviation of 2.000, suggesting more diverse views. In contrast, the nursing faculty showed unanimous, strong agreement ($M = 5.00$, $SD = 0.000$), while the School of Education had a slightly higher mean of 5.14, with minimal deviation ($SD = 0.690$), reflecting consistent strong agreement. Continuing and professional studies presented a mean of 3.00, with the highest variability ($SD = 2.828$), indicating neutral attitudes toward academic integrity and guidelines. Overall, the faculty members' ratings hovered around neutral to agreement on the need for specific

academic integrity guidelines in the age of AI (RQs 1 and 2), with a mean value of 6.00, representing strong agreement and 1.00 indicating strong disagreement.

Non-Significant Differences and Their Tendencies: In the analysis of Component 1 (i.e., AI Resources and Institutional Support), no significant differences were found concerning the faculty members' gender, age group, leadership role, and highest degree reached. Despite the lack of statistical significance, the data indicate a general tendency among faculty members to have attitudes ranging from neutral to moderate agreement on the need for AI resources and institutional support, with a mean rating of 4.16 and a standard deviation of 1.838. This indication suggests a moderately positive perception of the faculty for AI education, albeit with a relatively large degree of variation in their responses (RQ 1).

In evaluating Component 3 (i.e., Training on Academic Integrity in the Age of AI), the analysis did not find significant differences in faculty members' attitudes when segmented by gender, age group, leadership role, or the highest degree attained. The general tendency in attitudes towards this component ranged from disagreement to a neutral stance, as indicated by a mean score of 2.58 and a standard deviation of 2.009. These data suggest a homogeneity of perceptions about how the institution lacked standard ways of addressing academic integrity in the age of AI across various demographic and professional faculties' degrees (RQs 2 and 3).

In assessing Component 5 (i.e., Awareness of AI Benefits and Concerns and Academic Integrity Strategies), the analysis yielded no significant differences in faculty members' attitudes when segmented by gender, age group, leadership role, or highest degree obtained. Despite the lack of statistically significant variation across these demographics, the overall tendency among faculty members leaned towards an agreement, with the mean attitude rating close to the agreed range ($M = 4.79$) and a moderate level of dispersion in responses ($SD = 1.455$). This finding indicates that faculty and staff generally have a positive reception of the awareness of AI benefits, concerns, and strategies to ensure academic integrity (RQs 1 and 2).

Item-Based Correlation Analysis: While the sets of questions on university, college/school, and department/unit were highly correlated (i.e., Q1–18), responses were otherwise. Indeed, strong correlations were only observed in four pairs (i.e., 0.955 for Q8 and Q9, 0.970 for Q11 and Q10,

0.986 for Q14 and Q13, and 0.965 for Q15 and Q14) as highlighted in [Table A6](#) (in the [Appendix](#)), all centred on training for AI tools and guidelines for academic integrity. This finding reflects the inconsistency of guidelines on using AI tools in academic activities from various units (RQs 2 and 3). Interestingly, some extremely weak items to non-existing correlations were seen, as highlighted in [Table A7](#) (in the [Appendix](#)). It shows that for faculty and staff,

1. the clarity and currency of academic integrity guidelines provided by various units (Q10 and Q15) had little to do with people's understanding of the benefits of using AI tools in educational activities (Q19). The correlation values -0.002 and -0.015 revealed the lack of specific guidelines for AI tools.
2. the low correlation between people's support for restricting AI usage in academic activities (Q21) and whether clear guidelines were given for using AI tools (Q3) also suggests the need for specific guidelines. The low correlation value, -0.023 , showed that support for restricting AI is still individual and intrinsic at this stage.

Chi-Squared Tests for Independence

Findings of item-based analysis for the faculty and staff survey (see [Table A8](#) in the [Appendix](#)) indicate the following:

1. Gender and age groups showed no statistical significance in any questions, as no p -value was lower than .05, showing that other factors were more critical for faculty and staff.
2. Leadership was statistically significant ($p = .03$) regarding whether the department/unit offered training about academic integrity in educational activities. This shows the need for an organised/orchestrated movement that equips stakeholders with the critical use of AI.
3. The institution's role in using strategies for academic integrity was statistically significant ($p = .01$).
4. The highest degree and affiliated school/college/unit were the most statistically significant categories of responses, involving training and guidelines on AI tools and academic integrity, respectively.

Analysis of Survey Responses from Students

The student survey has been proven reliable, indicated by a robust Cronbach's alpha value of .920, reflecting a prominent level of internal consistency among the survey items. When standardising the items,

Cronbach's alpha slightly increased to .938, which further signified that the scale's items consistently measured the same underlying construct (see [Table A9](#) in the [Appendix](#)). This level of Cronbach's alpha suggests that the scale in the student survey was reliable for measuring whatever construct it was designed to assess, and the consistent results across all cases showed a robust set of items.

Despite this reliability, due to the low valid sample size ($N = 17$), the data from the survey may not be suitable for robust parametric methods such as factor analysis. Given this limitation, the survey data was analysed item by item using non-parametric exact solutions. These methods are more suitable for the data as they do not presuppose normal distribution and are better equipped to handle smaller sample sizes while offering valuable insights.

The items in descriptive statistics for the student survey (see [Table A10](#) in the [Appendix](#)) used the same scales as in descriptive statistics for the faculty and staff survey (see [Table A2](#) in the [Appendix](#)). However, student responses showed a different distribution from the survey for faculty and staff. Responses about AI institutional support and guidelines were skewed towards 'Disagree,' with scores below 2 consistently, while responses about academic integrity were between mid-2s and mid-3s. Understanding the benefits and concerns of AI tools, using strategies to uphold academic integrity in the AI era, and supporting the restriction of AI tools scored the highest (mid-3s to mid-5s).

Correlation: Aligned with the faculty and staff survey results, while the questions asked were highly correlated, responses were not (see [Table A11](#) in the [Appendix](#)). Strong correlations were only observed in six pairs (1.000 for Q14 and Q13; 1.000 for Q15 and Q13/Q14; 1.000 for Q17 and Q16; 0.996 for Q18 and Q16/Q17) as highlighted in [Table A6](#) (in the [Appendix](#)), all centred on guidelines and training for academic integrity. Meanwhile, weak correlations elsewhere suggest the incoherence of guidelines and training for GenAI tools from various units.

Interestingly, some extremely weak to non-existent correlations were observed, highlighted in [Table A12](#) (in the [Appendix](#)):

1. Students' understanding of the concerns about using GenAI tools in educational activities had little to do with whether the university provides clear or up-to-date guidelines.

- 2. Similarly, whether the university provided clear or up-to-date guidelines did not affect students’ strategies to ensure academic integrity in educational activities.

Chi-Squared Tests for Independence

Unlike the results of the survey analysis for faculty and staff, [Table A13](#) (in the [Appendix](#)) shows that the only statistically significant category for students survey was the age group. A close look at [Figure 2](#) shows that unlike the faculty and staff survey, where the Highest Degree and Affiliated Unit were the most crucial factors affecting responses to questions about training and guidelines on academic integrity and GenAI tools (Q9, Q11, Q18), age was the only determining factor for students (Q2–Q9).

In short, quantitative data from both surveys revealed that participants held varied attitudes towards GenAI in Education; perceived the effectiveness of the current AI guidelines and policies as neutral to disagreement (means at the 2s); perceived academic integrity as necessary (means at the upper 3s to 4s); held mixed feelings about AI benefits and strategies (means at the low to mid-3s), and were highly concerned about AI tools (means at 4.92). While data showed age group was the only significant difference for students, there was uniformity in perceptions from faculty and staff across gender and age. However, the highest degrees and

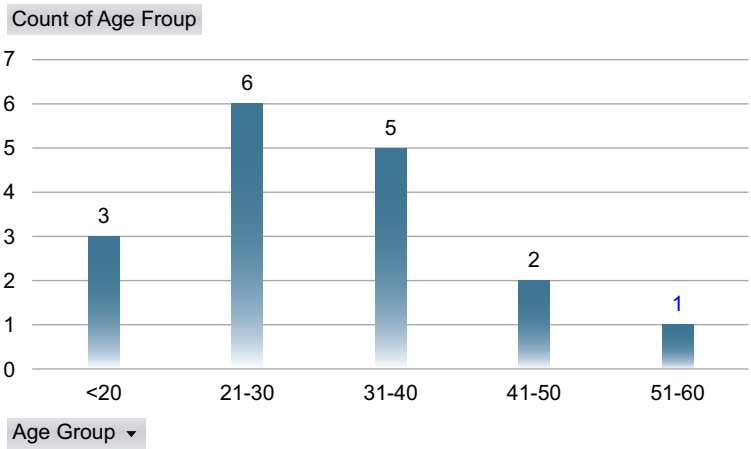


FIGURE 2. Count of Age Groups in the Student Survey

leadership roles in units and the institution were significant factors for faculty and staff in providing support, training, resources, and updated policies for AI and academic integrity. The results helped pinpoint specific areas needing updates and support.

Qualitative Results: Focus Groups, Survey Open-Ended Questions, and Par Process Results

Qualitative data analysis draws on information from the PAR process, focus groups, and open-ended survey questions from faculty/staff and students. Themes emerged around general AI education/training needs and the desire for specific university/departmental guidance. While most results advocated GenAI as an educational tool, some expressed concerns and even suggested banning it in educational settings.

What are Stakeholders' Perceptions and Knowledge of GenAI Tools in Teaching and Learning?

The stakeholders' perceptions and knowledge of GenAI tools in teaching and learning varied across dimensions. While most participants shared their interest in learning more about incorporating GenAI tools into their work, there was also much concern about how to use the technology in the classroom. The data gathered was divided into two sections related to the question: perceptions and knowledge.

Perceptions: There were mixed perspectives among stakeholders regarding GenAI tools. While some expressed excitement about GenAI affordances, such as its integration into learning and task optimisation, others hesitated about GenAI tool usage and advocated for restrictions to prevent compromising ethical issues and learning processes. Overall, faculty, staff, and students recognised GenAI tools as powerful. At the same time, they exhibited varying levels of concern about their usage. This sentiment was echoed by a stakeholder who viewed GenAI as a liberating force, likening it to calculators that freed individuals from manual computations, stating, 'It's freeing you up to do other things just like a calculator freed up the time and energy it takes to do computations.'

However, stakeholders also expressed concerns about equity, privacy, and bias issues associated with AI technology. They worried about algorithmic bias (i.e., unfair computer outputs) based on existing online data, highlighting the inherent biases in the data used to train AI systems. Faculty and staff raised concerns about the potential for 'hallucinations'

or inaccuracies in GenAI outputs, emphasising the need to address these issues to ensure fair and critical AI usage. Another concern raised was the impact of AI on students' learning development, particularly in diminishing the craft of writing and critical thinking skills. Additionally, there were concerns about specific GenAI tools being detrimental to students' cognitive abilities and the need for more consensus on how or when to credit AI in academic work, reflecting uncertainties surrounding pedagogical and academic integrity issues related to GenAI integration in education. One professor made the following case:

Some of the challenges that I have with AI is the information upon which it's being programmed, and it's being used, and I'm worried about the amount of what they refer to as hallucinations or algorithmic bias that comes from the data because of the inherent prone towards biases that it has because after all it is coded by humans and it is picking up data that we have out there in it's being coded on data that is sort of relatively open to the public. So, I am concerned about that. I am concerned about the biases of the data in terms of how it can be used.

Knowledge: Most stakeholders acknowledged the importance of implementing GenAI in classes to educate students on its real-world applications. However, students, professors, and staff recognised the need for guidelines, protocols, and broader frameworks to ensure the effective and responsible use of GenAI tools in education. Training and education for the university community on AI ethics were seen as crucial, as well as fostering conversations among faculty, students, and administrators about the usage of AI and how to set boundaries regarding using AI effectively. There was recognition that GenAI is becoming increasingly prevalent and that proactive measures are needed to navigate its integration while upholding academic integrity principles. One professor said, "Teaching students how to use AI responsibly because once they leave the institution. Yeah, there might be an expectation in the workplace. . . . So if students don't know the ins and outs of using AI to do a mockup, they're gonna be at a disadvantage entering the workforce."

What are Stakeholders' Perceptions of Academic Integrity in the AI Era?

A central theme amongst stakeholders acknowledged the necessity of redefining academic integrity, considering the integration of GenAI tools

into scholarly practices. Faculty, staff, and administrators underscored the necessity of reevaluating institutional core values integral to academic integrity, emphasising that the redefinition process should reflect the profound impact of AI on the educational environment, ‘we need to reassess our values and the metrics that we are examining, so that we are not penalizing resourceful students.’

Faculty, staff, and students expressed concerns about distinguishing between academic integrity and misconduct in AI usage, prompting discussions on ethical guidelines to uphold academic standards effectively. There was also an emphasis on the connection between academic integrity and honesty, stressing the importance of transparency in utilising GenAI tools. Staff, faculty, and administrators suggested that ethics should remain central in AI endeavours within the university’s values. Moreover, there was strong advocacy for integrating AI into educational settings while adapting assessment methods to evaluate output quality rather than penalising its utilisation. Some stakeholders suggested that academic integrity is about ‘authentic engagement with the content,’ valuing students who ‘take a chance to actually think and think creatively and bring ties from other classrooms [into what] they’ve learned.’

Unlike summative tests, authentic assessment involves evaluating students based on real-world tasks that reflect their understanding and application of knowledge rather than rote memorisation or formulaic responses. This approach was seen as more resilient to the misuse of GenAI tools. A faculty participant shared their reliance on authentic assessment: ‘I rely heavily on authentic assessment. So, an assessment that students are either taking from their experience or something that they’re going to use in their current careers.’ Authentic assessments can mitigate concerns about academic integrity by ensuring that the work produced is closely tied to the student’s experiences and future professional contexts, making it harder for AI to generate the responses on their behalf. Overall, the consensus underscored the need for thoughtful integration of GenAI into academic practices while keeping academic integrity standards and supporting all stakeholders.

Stakeholders’ discussion pointed to the necessity of facilitating the use of GenAI in a supportive role that enriches educational experiences and fosters a culture of integrity adapted to the new reality imposed by technology. A student’s anecdote on the practical application of GenAI in

an educational setting underscored this discussion point. The student described a class activity in which their professor had requested ChatGPT to generate an essay on a topic related to the class content. The students were then tasked to critically assess the AI-generated essay and identify its flaws and areas where it fell short. Not only did this exercise offer a hands-on opportunity to engage with AI technology, but it also fostered critical thinking and analytical skills. By evaluating the AI-produced content, the student learned to discern the strengths and weaknesses of AI, understanding its limitations and potential biases. Hence, this approach enriched the educational experience and instilled a culture of integrity, as the student was encouraged to engage with AI thoughtfully and critically, ensuring that they developed the necessary skills to use AI ethically and effectively in their academic pursuits.

How Effective are the Existing University Guidelines, Policies, Practices, and Strategies for Academic Integrity and GenAI Usage?

The research analysis pointed to the overall sentiment of disappointment among stakeholders concerning the extent of support and guidance available to deliver and integrate GenAI tools. Some professors and staff expressed that this disappointment was not attributed to the capabilities of GenAI tools but rather to the perceived inadequacies in the ecosystem surrounding these tools, including institutional support structures, user training, and allocation of technological resources. One faculty member argued, ‘There’s been no cohesion across my academic unit, much less further into the University. I put a tremendous amount of effort into seeking out those opportunities. Talking with other people about these things is important. I’ve met with many people in this focus group and worked with them in different ways, but the University doesn’t pick it up.’

Stakeholders articulated a sense of being left to navigate the complexities of AI integration with inadequate guidance and training, leading to a scenario in which the potential benefits of AI could be more leveraged. Professors and librarians argued that this lack of guidance hindered improving educational outcomes and innovating teaching methods, which was a missed opportunity for the institution to leverage AI’s transformative capabilities. One professor commented, ‘Faculty were left with zero guidance. Right? Just as support staff were also left with zero guidance.’

Another emerging theme from the data signified a collective call among participants for more significant innovation in using and developing GenAI tools in educational spheres, coupled with strong advocacy for inclusive involvement in related policymaking adaptations. One typical comment was, ‘Who’s making the decisions? Also, notice that there’s not anyone from . . . like the Office of Online Learning on the AI Committee.’

Stakeholders underscored the necessity for an institutional environment that encourages faculty and student participation in ideating and creating policies governing GenAI tool deployment. This call for involvement was predicated on believing that such collaborative approaches can lead to more innovative, relevant, and responsive AI solutions that better align with learners’ and educators’ needs and expectations. Moreover, stakeholders contended that direct involvement in policy formation would democratise AI application in education and adequately address ethical considerations and equity concerns. Collectively, these themes elucidated a dynamic interface between stakeholders and GenAI tools in education, characterised by a desire for improved support mechanisms and a more participatory model of innovation.

What Actions are Needed to Prepare the Institution for the Age of AI?

Thematic analysis of the data suggested a pressing concerted call for cohesive efforts toward constructive AI education and integration. This would foster an inclusive educational ecosystem where AI is taught and integrated into practical experience, encouraging a hands-on learning approach. The approach advocated for bridging the gap between theoretical knowledge and practical application, ensuring stakeholders are not just passive learners but active participants in the discourse of AI. Furthermore, there was an emphasis on prioritising education over legislation as an emerging theme. This approach emphasises the importance of educating about AI before creating regulations: ‘If we can understand and demystify this tool, we can begin to use it effectively.’ Furthermore, this finding indicates that cross-sector leadership and support are crucial for advancing AI literacy initiatives.

Finally, the findings showed that revising existing policies and providing clear expectations and guidelines is essential. Stakeholders emphasised the need for a comprehensive evaluation of AI-related policies to ensure their flexibility and responsiveness to the constant advancement

of AI technology, which one stakeholder stated should be a ‘cycle of continuous improvement regarding AI: learn, revisit and research the best uses of AI, train and disseminate, and reflect.’ Stakeholders also highlighted the importance of offering clear, accessible, and actionable directives and guiding AI’s ethical, informed, and judicious usage.

PAR Process: Comparison and Interpretation

The PAR members’ brainstorming activity on the issues echoed similar themes from the qualitative and quantitative results. Among the major themes were excitement and concerns about the new opportunities and challenges, eagerness to learn new GenAI tools, and the need for clear guidelines. Regarding ways to improve the AI education environment, integrating AI in curriculum and assignments and access to support came up. On the list of priorities for action were professional development on using GenAI tools, CAIL workshops, and inclusive collaboration in making guidelines and policies for faculty and students.

To conclude, the results from the multiple data sets, quantitative, qualitative, and PAR analysis triangulate with each other and show important significant themes and trends in findings. Stakeholders showed diverse perceptions and levels of knowledge regarding GenAI tools in the educational context. Some showed a readiness to integrate AI into their pedagogy, perceiving these tools as beneficial for enhancing teaching and learning experiences. In contrast, others expressed reservations about potential challenges, such as concerns about academic integrity. Many stakeholders, recognising the significance of academic integrity, asked for clear guidelines, policies, support, and training for GenAI. Based on the results, the PAR team conducted a SWOT analysis ([Mind-Tools, n.d.](#); see [Table 3](#)). The analysis categorises the findings into strengths, weaknesses, opportunities, and threats in facing the excitement and concerns of GenAI tools and academic integrity and guides the recommendations for actions.

DISCUSSION AND CONCLUSION

The study results are snapshots of one HEI in the changing landscape of GenAI. They echo themes from existing literature and reveal current trends and thoughts, illustrating the innovations and concerns within GenAI ([Bahroun et al., 2023](#)). This section discusses the multifaceted impact of

TABLE 3. Adapted Strengths, Weaknesses, Opportunities, and Threats Analysis

Strengths
• Availability of faculty, librarians, and IT support with AI expertise • Student body is eager to learn and explore new technologies • Awareness of AI and its potential impact on education • Commitment to preparing students with the skills they need in the AI-driven world • Awareness of the need for the development of AI ethical practices • Recognition of AI issues and a willingness to address them • AI potential to optimise and transform some research, learning, and teaching practices
Weaknesses
• AI technology outpaces comprehensive research and guidelines • Internal workings and potential biases of AI systems are not widely known • Unclear guidelines and regulations for responsible AI use • Fear of the unknown and resistance to change hinder widespread AI adoption • Balancing academic integrity with the rise of GenAI tools • The CAIL framework and training are still under development • Lack of evidence-based knowledge to guide AI use in teaching, learning, and research
Opportunities
• Revision of guidelines and frameworks to address AI use and ensure ethical learning • AI's potential to optimise and transform research, learning, and teaching practices • Coordinated efforts across departments to leverage AI for transformative learning • Fostering a collaborative culture of responsible AI practices through open discussion, knowledge sharing, and interest groups. • Integrating critical AI education into programmes, preparing students to thrive in the AI-powered world. • Cultivating a strong AI community through ongoing AI open discussion, knowledge sharing, and interest groups.

(Continued)

TABLE 3. Continued

Threats
• Ethical concerns about data privacy, inherent biases, and misinformation • Plagiarism and copyright infringement due to AI misuse • Mixed views on AI can hinder the development of critical thinking and problem-solving skills necessary in an AI-driven world • Lack of transparency in AI algorithms • Insufficient funding for AI research and initiatives can hinder responsible development and adoption. • Undemocratic AI access exacerbates existing educational disparities • Punitive measures for AI use can discourage open discussion and exploration of AI literacy.

Source: [Mind-Tools \(n.d.\)](#).

Note: IT = information technology.

these findings by analysing HEI’s strengths and opportunities, followed by weaknesses and threats in handling AI. The researchers now propose actionable interventions to safeguard academic integrity and ensure ethical GenAI implementation at the institution.

Strengths and Opportunities: GenAI Tools Enhance Social Experience
Aligned with emerging GenAI policies ([Chan, 2023](#); [UNESCO, 2023](#)), interdisciplinary engagement through a human-centred approach is recommended to enact effective and ethical practices with GenAI tools. This research involved members of the PAR team and multiple university stakeholders from diverse departments whose active engagement showed their willingness to share and exchange information related to GenAI. The observed evolution of opinions as participants interacted through focus groups and PAR discussions underscores the importance of collaborative policymaking in navigating the ethical use of GenAI.

Including students, faculty, staff, and administrators in this study, they fostered a comprehensive understanding of GenAI challenges across campus, echoing the three interconnected dimensions of [Chan’s \(2023\)](#) AI Ecological Education Policy Framework: Pedagogical, Operational, and Governance. While Chan’s model highlights the role of information technology staff in the Operational dimension, this collaborative project emphasises the importance of librarians and instructional designers in

institutional AI implementation. In addition, the PAR investigation actively involved students in surveys, focus groups, and the research team itself, enhancing often poorly represented voices within AI decision-making policies (Sullivan et al., 2023). Indeed, students proved a true asset for this discussion, contributing tremendously to making sense of the complexities inherent to GenAI technology.

Stakeholders showed diverse levels of experience with GenAI tools. However, they mostly agreed that if ethical considerations are in place, GenAI can reshape education and make teaching more effective, learning more engaging, and writing more dynamic (Bahroun et al., 2023; Currie, 2023; UNESCO, 2023). Results showed the need for critical GenAI training and ethical considerations in establishing GenAI guidelines and policies related to campus activities and classwork. Stakeholders voiced that AI should be integrated into all programmes so universities can equip the students, as the future workforce, with ethical, innovative, and real-world GenAI applications. These findings are aligned with Schiff (2022) about the role of AI in education (AIED) policy discourse. Although AI policy discourse often marginalises the transformative domain of AIED, this study underscores the importance of innovations and critical thinking when co-constructing AI ethical uses.

Reconceptualising assessments and assignment outcomes due to GenAI implications is a prominent subject of research (Birks & Clare, 2023; Cotton et al., 2023; Moorhouse et al., 2023). UNESCO (2023) recommends prioritising higher-order thinking outcomes to effectively integrate ethical uses of GenAI applications without compromising students' learning development. A student participant had a class assignment in the honours programme to critique GenAI-generated content by analysing a GenAI essay on a foundational topic pertinent to the class. The activity allowed students to identify both strengths and gaps in GenAI-generated outputs. The preceding pedagogical strategy revealed an interesting higher-order thinking activity and an effective way to have students use GenAI through 'embodied interaction' (Long & Magerko, 2020, p. 5) to accomplish class assignments.

Engaging students in higher-order thinking tasks should discourage their chances of academic misconduct, such as plagiarism when using GenAI tools (UNESCO, 2023). While pedagogical activities like prompting students to reflect on their learning experiences or collaborating with

peers on projects have been suggested, they require additional teacher guidance. The need for empirical evidence about teaching and learning with AI suggests that stakeholders should collaboratively develop strategies that identify gaps in GenAI-generated work and adjust student assessments to prevent academic misconduct.

Stakeholders supported educative measures as the centre of AI policy because academic integrity is a cornerstone for scholarly research, education, and the fundamental values of educational institutions. They called for implementing thorough professional development for faculty and students, pushing against following a set of rules acritically. The stakeholders agreed that students should be taught ‘contextually specific constructs of text ownership’ (Anson, 2022, p. 37) to understand the social conventions of different productions better. In the context of numerous GenAI tools and the absence of CAIL, it is imperative that educators, scholars, institutions, and students collectively engage in open conversations about academic integrity in the age of GenAI.

Weaknesses and Threats: The Absence of GenAI Support Leads to Uncertainty

Stakeholders’ lack of training in CAIL and the proliferation of GenAI tools have overwhelmed faculty, staff, and students. Regardless of expertise, many stakeholders were concerned about academic integrity and uncertainty regarding GenAI technologies, which overshadowed the critical moment to educate students and to benefit from GenAI. In addition, the lack of institutional support and guidance due to uncertainty discouraged some stakeholders from exploring GenAI technologies and including them in their practices and assignments. Stakeholders also voiced GenAI concerns on privacy, bias, transparency, and access issues. These considerations resonate with GenAI issues in the literature (Long & Margerko, 2020; UNESCO, 2023), and the study results call for empirical evidence on effective strategies to mitigate these GenAI issues.

Participants believed institutional policies should not be punitive. In addition, participants said that policies should be adapted to the ever-evolving nature of GenAI tools. Thus, De Maio’s (2024) framework, with flexibility and currency on top of access, approach, responsibility, detail, and support, can effectively help institutions and faculty update their GenAI policies and mitigate academic integrity issues.

There is a clear call for training in the critical use of GenAI in education and preparing students for jobs in the new AI landscape. Although GenAI does not negate the need for critical thinking, it may require new pedagogical approaches to teach it (Park et al., 2021). This skill is conscientisation (Freire, 2000), which locates subjects politically, socially, and economically in the world, empowering them to work towards improving and transforming their communities. In education, fostering critical thinking abilities is essential for students to analyse information, evaluate arguments, and make informed decisions.

Actions Plans

The PAR team is committed to social action and transformation. Whether the plan is to legislate or educate, the findings suggest that an approach with CAIL education and guidelines for AI use is more likely to establish academic integrity in the age of AI. Considering the results oppose blanket GenAI bans, HEI stakeholders must establish AI policies, guidelines, and critical AI education through a human-centred approach (UNESCO, 2023). This combination ensures academic integrity while stakeholders learn how AI can optimise and transform teaching and learning experiences.

Inspired by Chan's (2023) AI Ecological Education Policy Framework and the PAR findings, the team determined a research-based action plan with the following priorities: CAIL education for the community, professional development to integrate GenAI tools for faculty, and the establishment of clear guidelines and policies to embrace the use of GenAI tools and uphold academic integrity.

The Operation Dimension

To inform everyone about the benefits of new GenAI tools, embrace innovative technology, and use it ethically, the researchers recommend HEIs provide CAIL training across all stakeholder groups. With the newly updated model of CAIL introduced here, audiences can learn to *access* GenAI tools, critically *analyse* various GenAI tools and their products, *evaluate* the quality and credibility of big data and AI non-predictability, know when to *apply* human intervention to avoid machine biases, *create* content ethically and appropriately, reflect on one's conduct for social responsibility, and *act* to create a critical AI-literate environment.

The Pedagogical Dimension

The researchers recommend professional development (PD) for faculty and staff on integrating GenAI tools in teaching, learning, and research. PD focuses on accessing GenAI tools, integrating them into curriculum, instruction, and assignments, teaching students CAIL, and preparing students for the AI workforce.

The Governance Dimension

The researchers recommend establishing clear guidelines and policies about using GenAI tools while upholding academic integrity. This process must be inclusive and collaborative, with voices from administration, technology experts, staff, faculty, and students.

Implications and Limitations

An essential contribution of this study is the collective effort to reinvent the existing CML and AI models (Chan, 2023; Long & Magerko, 2020; UNESCO, 2023; Wan, 2010; Wan et al., 2015) and conceptualise a CAIL education framework. CAIL addresses different skills imposed by AI technology through critical lenses, capturing social responsibility and broader political implications of GenAI technology, such as environmental impact, exploitation of workers, labour displacement, data misrepresentation, and many others. This new theoretical framework guided this research project and will be improved as the research team puts CAIL into practice.

With institutional support, the previously described action plan could be implemented immediately. The recommendations align with the mission of the home institution where the study took place. The age of AI presents opportunities to look for pathways to recommit to what stakeholders see as the most foundational and impactful components of education. The shared vision of knowledge engagement is based on social justice, which prompts individuals to reflect and orient themselves in response to changing contexts. Furthermore, it compels committed educators to focus on relationships, learning about oneself and how one's unique gifts can contribute to the greater good, and orienting learning toward questions of how to create a more just world. Integrating CAIL would uphold these values and advance the institution's commitment to academic integrity in this new age of AI.

In addition, the three recommendations drew upon the three dimensions of the AI Ecological Education Policy Framework (Chan, 2023) because its intersectoral nature allows for a more comprehensive AI implementation. The informed action plan will be shared with colleagues and leadership at the authors' home institution and with other HEIs through presentations and publications. The action plan will help HEIs take advantage of AI development and meet its challenges better. The researchers encourage HEIs in similar situations to borrow our action plans and the CAIL model, learn from our AI learning experience, and share theirs. Higher education institutions may duplicate the action plans, test the CAIL model in different contexts, and help improve the emerging model.

During the study process, the team realised there were some limitations. First, the research project received fewer than optimal survey responses despite the numerous efforts to call for participation. The low response rates could be due to the timing of the survey distribution, which coincided with a university break, the discomfort in discussing academic integrity, uncertainty about AI, and an attitude such as 'I don't know anything about AI.' To enhance participation rates in the future, researchers will investigate alternative recruitment methods and scheduling. Second, the survey response rates are not available, as the populations the team reached out to are not available. Snowball sampling could be the primary reason, as the researchers asked the university administration to email the calls, post them in newsletters, and announce them at various events. Finally, as AI is advancing daily with emerging new tools, new research, and publications, this PAR study represents snapshots of what the researchers see at this moment.

CONCLUSION

This mixed-method PAR research, conducted with university stakeholders, offers a prompt exploration of academic integrity in the age of AI. The findings, pointing towards an approach that guides and educates people about AI, led to a research-based action plan for one HEI to navigate the multifaceted impacts of GenAI. This plan, however, holds broader applicability. Other institutions can localise and build on it to address the opportunities and challenges of AI in education.

Future research will evaluate the implementation of this plan and GenAI's long-term impact on academic integrity and teaching and learning.

Furthermore, the emerging framework for CAIL education will be further tested, refined, and disseminated to help educators and learners navigate AI's evolving landscape in education. By prioritising responsible and critical AI use, newer technology can foster plural voices and creativity, fully transforming teaching and learning practices.

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APPENDIX

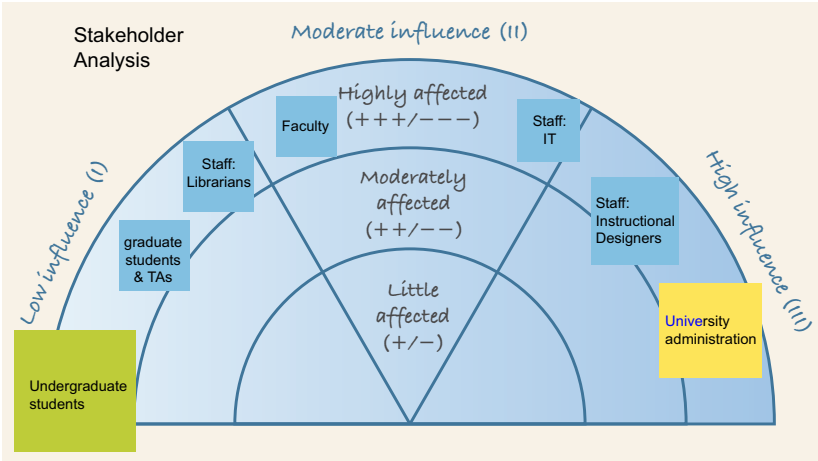


FIGURE A1. Stakeholder Rainbow Analysis (Chevalier & Buckles, 2019, p. 249)

TABLE A1. Original Survey Items: Artificial Intelligence Tools and Academic Integrity

Part A: Likert Scale Questions

A. Rate each unit in the Table and whether it has provided clear guidelines for using artificial intelligence tools in educational activities at the University.

	Strongly Disagree 1	Disagree 2	Slightly Disagree 3	Slightly Agree 4	Agree 5	Strongly Agree 6	NA or I don't know 0
Q1. University							
Q2. College/ School							
Q3. Department/ Unit							

(Continued)

TABLE A1. Continued

B. Rate each unit in the Table and whether it has provided up-to-date guidelines for using artificial intelligence tools in educational activities at the University.

	Strongly Disagree 1	Disagree 2	Slightly Disagree 3	Slightly Agree 4	Agree 5	Strongly Agree 6	NA or I don't know 0
--	------------------------	------------	------------------------	---------------------	---------	---------------------	----------------------------

Q4.
University
Q5. College/
School
Q6.
Department/
Unit

C. Rate the unit in the Table to indicate whether it offers training on using artificial intelligence tools in educational activities at the University.

	Strongly Disagree 1	Disagree 2	Slightly Disagree 3	Slightly Agree 4	Agree 5	Strongly Agree 6	NA or I don't know 0
--	------------------------	------------	------------------------	---------------------	---------	---------------------	----------------------------

Q7.
University
Q8. College/
School
Q9.
Department/
Unit

(Continued)

TABLE A1. Continued

D. Rate the unit in the Table to indicate whether it has clear guidelines for academic integrity in educational activities at the University.

	Strongly Disagree 1	Disagree 2	Slightly Disagree 3	Slightly Agree 4	Agree 5	Strongly Agree 6	NA or I don't know 0
--	------------------------	------------	------------------------	---------------------	---------	---------------------	----------------------------

Q10.

University

Q11.

**College/
School**

Q12.

**Department/
Unit**

E. Rate the unit in the Table to indicate whether it has up-to-date guidelines for academic integrity in educational activities at the University.

	Strongly Disagree 1	Disagree 2	Slightly Disagree 3	Slightly Agree 4	Agree 5	Strongly Agree 6	NA or I don't know 0
--	------------------------	------------	------------------------	---------------------	---------	---------------------	----------------------------

Q13.

University

Q14.

**College/
School**

Q15.

**Department/
Unit**

(Continued)

TABLE A1. Continued

F. Rate the unit in the Table to determine whether it offers training regarding academic integrity in educational activities at the University.

	Strongly Disagree 1	Disagree 2	Slightly Disagree 3	Slightly Agree 4	Agree 5	Strongly Agree 6	NA or I don't know 0
--	------------------------	------------	------------------------	---------------------	---------	---------------------	----------------------------

Q16.
University

Q17.
College/
School

Q18.
Department/
Unit

G. Rate each of the following statements based on your level of agreement.

	Strongly Disagree 1	Disagree 2	Slightly Disagree 3	Slightly Agree 4	Agree 5	Strongly Agree 6	NA or I don't know 0
--	------------------------	------------	------------------------	---------------------	---------	---------------------	----------------------------

Q19. You
understand
the benefits
of using
artificial
intelligence
tools in
educational
activities at
the
University

Q20. You
understand
the
concerns
about using

(Continued)

TABLE A1. Continued

	Strongly Disagree 1	Disagree 2	Slightly Disagree 3	Slightly Agree 4	Agree 5	Strongly Agree 6	NA or I don't know 0
artificial intelligence tools in educational activities at the University.							
Q21. You support restricting the use of artificial intelligence tools in educational activities at the University.							
Q22. You use strategies to embrace artificial intelligence tools in educational activities at the University.							
Q23. You use strategies to ensure academic integrity in educational activities at the University.							

TABLE A1. Continued

Part B: Open-ended Questions

- Q24. In your opinion, how should the University respond to artificial intelligence and academic integrity?**
- Q25. Share any recommendations you have regarding artificial intelligence and academic integrity at the University.**

Part C: Demographics**Q26. What is your role at the University?**

- a. Undergraduate Student
- b. Graduate Student
- c. Faculty
- d. Staff
- e. Administrator

Q27. Gender:

- a. Female
- b. Male
- c. Non-binary
- d. Third gender
- e. Prefer not to say
- f. Other

Q28. Age Group:

- a. 20 or under
- b. 21–30
- c. 31–40
- d. 41–50
- e. 51–60
- f. 61–70
- g. Over 70

Q29. Are you in a leadership role at the University? (e.g., organizations for students, faculty, or staff)

- a. Yes
- b. No

(Continued)

TABLE A1. Continued

Q30. What is your affiliated school/college/unit?		
a. Arts and Sciences		
b. College of Business		
c. School of Communication		
d. School of Continuing and Professional Studies		
e. School of Education		
f. School of Environmental Sustainability		
g. School of Health Sciences and Public Health		
h. School of Law		
i. School of Medicine		
j. School of Nursing		
k. School of Social Work		
l. My affiliated unit is not listed above.		
Q31. What's your highest degree?		
a. High school		
b. Associate's		
c. Bachelor's		
d. Master's		
e. Doctoral		
f. My highest degree is not listed above		

TABLE A2. Analysis of Internal Consistency of the Faculty and Staff Survey

Cronbach's alpha	Cronbach's alpha based on standardised items	N of items
.881	.890	23

TABLE A3. Descriptive Statistics for Faculty and Staff Survey

Statements based on the Likert scale survey items in Appendix Table A1							
	N	Mean		Std. deviation	Skewness		Kurtosis
		Statistic	Std.		Statistic	Std. error	
Q1 Whether the University has provided clear guidelines for using artificial intelligence tools in educational activities.	38	2.66	1.665	0.172	0.383	-0.527	0.750
Q2 Whether the College/School has provided clear guidelines for using artificial intelligence tools in educational activities.	38	1.95	1.610	0.582	0.383	-0.229	0.750
Q3 Whether the Department/Unit has provided clear guidelines for using artificial intelligence tools in educational activities.	38	2.24	1.635	0.380	0.383	-0.747	0.750
Q4 Whether the University has provided up-to-date guidelines for using artificial intelligence tools in educational activities.	38	2.21	1.527	0.441	0.383	-0.137	0.750
Q5 Whether the College/School has provided up-to-date guidelines for using artificial intelligence tools in educational activities.	38	1.79	1.379	0.923	0.383	1.428	0.750
Q6 Whether the Department/Unit has provided up-to-date guidelines for using artificial intelligence tools in educational activities.	38	2.08	1.600	0.700	0.383	-0.179	0.750
Q7 Whether the University offers training for using artificial intelligence tools in educational activities.	38	2.11	1.689	0.430	0.383	-0.791	0.750

(Continued)

TABLE A3. Continued

Statements based on the Likert scale survey items in Appendix Table A1	N	Mean		Std. deviation	Skewness		Kurtosis	
		Statistic	Std. error		Statistic	Std. error	Statistic	Std. error
Q8 Whether the College/School offers training for using artificial intelligence tools in educational activities.	38	1.87	1.545		0.835	0.383	0.246	0.750
Q9 Whether the Department/Unit offers training on using artificial intelligence tools in educational activities	38	2.16	1.685		0.847	0.383	-0.182	0.750
Q10 Whether the University has clear guidelines for academic integrity in educational activities.	38	4.16	1.838		-0.824	0.383	-0.408	0.750
Q11 Whether the College/School has clear guidelines for academic integrity in educational activities.	38	3.34	2.245		-0.378	0.383	-1.353	0.750
Q12 Whether the Department/Unit has clear guidelines for academic integrity in educational activities.	38	3.53	2.263		-0.465	0.383	-1.350	0.750
Q13 Whether the University has up-to-date guidelines for academic integrity in educational activities.	38	3.26	1.796		-0.536	0.383	-0.710	0.750
Q14 Whether the College/School has up-to-date guidelines for academic integrity in educational activities.	38	2.92	1.978		-0.261	0.383	-1.247	0.750
Q15 Whether the Department/Unit has up-to-date guidelines for academic integrity in educational activities.	38	3.05	1.931		-0.388	0.383	-1.117	0.750
Q16 Whether the University offers training regarding academic integrity in educational activities.	38	2.58	2.009		0.026	0.383	-1.321	0.750

(Continued)

TABLE A3. Continued

Statements based on the Likert scale survey items in Appendix Table A1									
	N	Mean		Std. deviation		Skewness		Kurtosis	
		Statistic	Std. error	Statistic	Std. error	Statistic	Std. error	Statistic	Std. error
Q17 Whether the College/School offers training regarding academic integrity in educational activities.	38	2.34	1.963	0.329	0.383	-1.026	0.750		
Q18 Whether the Department/Unit offers training regarding academic integrity in educational activities.	38	2.34	2.017	0.483	0.383	-0.974	0.750		
Q19 Whether the respondent understands the benefits of using artificial intelligence tools in educational activities.	38	3.76	1.777	-0.597	0.383	-0.684	0.750		
Q20 Whether the respondent understands the concerns about using artificial intelligence tools in educational activities.	38	4.79	1.455	-1.780	0.383	3.079	0.750		
Q21 Whether the respondent supports restricting the use of artificial intelligence tools in educational activities.	38	3.55	2.076	-0.471	0.383	-1.084	0.750		
Q22 Whether the respondent uses strategies to embrace artificial intelligence tools in educational activities.	38	3.21	1.862	-0.033	0.383	-1.245	0.750		
Q23 Whether the respondent uses strategies to ensure academic integrity in educational activities.	38	4.45	1.996	-1.414	0.383	0.679	0.750		
Valid N (listwise)	38								

TABLE A4. Kaiser–Meyer–Olkin measure and Bartlett’s test

Kaiser–Meyer–Olkin measure of sampling adequacy		0.660
Bartlett’s test of sphericity	Approx. chi-squared	802.420
	<i>df</i>	253
	Sig.	0.000

Source: [Shrestha \(2021\)](#).

TABLE A5. Principal Component Analysis: Total Variance Explained

Comp.	Total variance explained								
	Initial eigenvalues			Extraction sums of squared loadings			Rotation sums of squared loadings		
	Total	% of variance	Cumulative %	Total	% of variance	Cumulative %	Total	% of variance	Cumulative %
1	7.499	32.605	32.605	7.499	32.605	32.605	5.166	22.461	22.461
2	3.975	17.281	49.886	3.975	17.281	49.886	3.448	14.992	37.453
3	2.144	9.322	59.208	2.144	9.322	59.208	2.757	11.986	49.439
4	1.670	7.260	66.468	1.670	7.260	66.468	2.306	10.026	59.465
5	1.464	6.364	72.832	1.464	6.364	72.832	2.289	9.954	69.419

TABLE A6. Spearman’s Correlation Matrix for Questions with Correlation Values Greater than 0.95

	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15
Q7	1.000								
Q8	0.686	1.000							
Q9	0.559	0.955	1.000						
Q10	0.307	0.071	0.037	1.000					
Q11	0.467	0.127	0.134	0.970	1.000				
Q12	0.306	0.136	0.216	0.809	0.937	1.000			
Q13	0.373	0.225	0.099	0.746	0.733	0.663	1.000		
Q14	0.443	0.279	0.185	0.702	0.735	0.747	0.986	1.000	
Q15	0.416	0.282	0.329	0.614	0.692	0.761	0.896	0.965	1.000

Note: Data in bold indicate strong correlations.

TABLE A7. Spearman’s Correlation Matrix for Questions with Correlation Values Between −0.05 and 0.05

	Q3	Q9	Q10	Q15	Q16	Q19	Q21
Q3	1.000						
Q9	0.388	1.000					
Q10	0.185	0.037	1.000				
Q15	0.554	0.329	0.614	1.000			
Q16	0.011	0.174	0.562	0.527	1.000		
Q19	0.119	0.543	−0.002	−0.015	0.349	1.000	
Q21	−0.023	−0.577	0.177	0.267	−0.281	−0.572	1.000

Note: Data in bold indicate weak correlations.

TABLE A8. Item-Based Analysis: Matrix of P-values for Faculty and Staff Survey

	Role/ position	Leadership or not?	Gender	Highest degree?	Affiliated school/ college/unit	Age Group
Q1	0.800	0.470	0.520	0.320	0.390	0.770
Q2	0.390	0.660	0.390	0.830	0.650	0.480
Q3	0.070	0.150	0.760	0.800	0.570	0.690
Q4	0.470	0.690	0.390	0.700	0.810	0.670
Q5	0.060	1.000	0.750	1.000	0.930	0.780
Q6	0.470	0.230	0.810	0.700	0.870	0.210
Q7	0.720	1.000	0.130	1.000	0.790	0.810
Q8	0.850	1.000	0.060	0.550	0.380	0.520
Q9	0.250	0.690	0.190	0.040	0.040	0.160
Q10	0.250	1.000	0.070	0.490	0.150	0.270
Q11	0.020	0.740	0.460	0.050	0.000	0.420
Q12	0.130	0.740	0.460	0.210	0.090	0.300
Q13	0.110	1.000	0.770	0.640	0.170	0.850
Q14	0.200	1.000	0.810	0.530	0.150	0.680
Q15	0.450	1.000	0.740	0.640	0.410	0.690
Q16	0.750	1.000	0.660	0.770	0.020	0.200
Q17	0.800	1.000	0.450	0.320	0.520	0.630
Q18	0.470	0.030	0.240	0.010	0.390	0.320
Q19	0.230	1.000	0.250	0.450	0.780	0.600
Q20	0.850	1.000	0.790	0.270	0.860	0.510
Q21	0.460	1.000	0.120	1.000	0.420	0.930
Q22	0.340	0.520	0.460	0.370	0.100	0.100
Q23	0.010	0.090	0.200	0.830	0.860	0.650

Note: Data in bold indicates those with p-value smaller than 0.05 = significant.

TABLE A9. Analysis of Internal Consistency (Internal Consistency is Shown)

Cronbach's alpha	Cronbach's alpha based on standardised items	N of Items
.920	.938	23

TABLE A10. Descriptive Statistics for Student Survey

Descriptive Statistics									
Statements based on the Likert scale survey items in Appendix A1	N	Mean		Std. deviation		Skewness		Kurtosis	
		Statistic	Std. error	Statistic	Std. error	Statistic	Std. error	Statistic	Std. error
Q1 Whether the University has provided clear guidelines for using artificial intelligence tools in educational activities.	17	1.94		1.391		1.698		3.881	
Q2 Whether the College/School has provided clear guidelines for using artificial intelligence tools in educational activities.	17	1.82		1.286		2.167		6.990	
Q3 Whether the Department/Unit has provided clear guidelines for using artificial intelligence tools in educational activities.	17	1.94		1.298		1.872		5.723	
Q4 Whether the University has provided up-to-date guidelines for using artificial intelligence tools in educational activities.	17	1.76		1.348		2.047		5.699	
Q5 Whether the College/School has provided up-to-date guidelines for using artificial intelligence tools in educational activities.	17	1.76		1.300		2.233		7.048	
Q6 Whether the Department/Unit has provided up-to-date guidelines for using artificial intelligence tools in educational activities.	17	1.71		1.359		2.133		5.871	
Q7 Whether the University offers training for using artificial intelligence tools in educational activities.	17	1.59		1.278		2.732		9.551	

(Continued)

TABLE A10. Continued

Descriptive Statistics									
Statements based on the Likert scale survey items in Appendix A1	N	Mean		Std. deviation		Skewness		Kurtosis	
		Statistic	Std. error	Statistic	Std. error	Statistic	Std. error	Statistic	Std. error
Q8 Whether the College/School offers training for using artificial intelligence tools in educational activities.	17	1.53		1.281		2.861	0.550	10.093	1.063
Q9 Whether the Department/Unit offers training on using artificial intelligence tools in educational activities.	17	1.59		1.326		2.521	0.550	7.858	1.063
Q10 Whether the University has clear guidelines for academic integrity in educational activities.	17	3.29		1.929		-0.177	0.550	-1.347	1.063
Q11 Whether the College/School has clear guidelines for academic integrity in educational activities.	17	3.18		1.811		-0.225	0.550	-1.248	1.063
Q12 Whether the Department/Unit has clear guidelines for academic integrity in educational activities.	17	3.18		1.741		-0.306	0.550	-0.958	1.063
Q13 Whether the University has up-to-date guidelines for academic integrity in educational activities.	17	2.65		1.902		0.021	0.550	-1.101	1.063
Q14 Whether the College/School has up-to-date guidelines for academic integrity in educational activities.	17	2.47		1.807		0.198	0.550	-0.763	1.063
Q15 Whether the Department/Unit has up-to-date guidelines for academic integrity in educational activities.	17	2.47		1.807		0.198	0.550	-0.763	1.063
Q16 Whether the University offers training regarding academic integrity in educational activities.	17	2.47		1.940		0.632	0.550	-1.007	1.063

(Continued)

TABLE A10. Continued

Descriptive Statistics							
Statements based on the Likert scale survey items in Appendix A1	N	Mean		Std. deviation	Skewness		Kurtosis
		Statistic	Std. error		Statistic	Std. error	
Q17 Whether the College/School offers training regarding academic integrity in educational activities.	17	2.18	0.550	1.912	0.871	0.550	-0.388
Q18 Whether the Department/Unit offers training regarding academic integrity in educational activities.	17	2.06	0.550	1.784	1.023	0.550	0.392
Q19 Whether the respondent understands the benefits of using artificial intelligence tools in educational activities.	17	4.35	0.550	1.967	-1.064	0.550	-0.011
Q20 Whether the respondent understands the concerns about using artificial intelligence tools in educational activities.	17	5.00	0.550	1.323	-1.836	0.550	4.313
Q21 Whether the respondent supports restricting the use of artificial intelligence tools in educational activities.	17	3.47	0.550	1.807	0.126	0.550	-1.589
Q22 Whether the respondent uses strategies to embrace artificial intelligence tools in educational activities.	17	3.65	0.550	2.262	-0.493	0.550	-1.345
Q23 Whether the respondent uses strategies to ensure academic integrity in educational activities	17	4.47	0.550	2.239	-1.508	0.550	0.843
Valid N (listwise)	17						

TABLE A11. Spearman's Correlation Matrix for Questions with Correlation Values Greater than 0.95

	Q13	Q14	Q15	Q16	Q17	Q18
Q13	1.000					
Q14	1.000	1.000				
Q15	1.000	1.000	1.000			
Q16	0.754	0.638	0.638	1.000		
Q17	0.754	0.754	0.754	1.000	1.000	
Q18	0.759	0.759	0.759	0.996	0.996	1.000

Note: Data in bold indicate figures greater and 0.95 with strong correlation.

TABLE A12. Spearman's Correlation Matrix for Questions with Correlation Values Between -0.05 and 0.05

	Q1	Q5	Q10	Q13	Q20	Q23
Q1	1.000					
Q5	0.738	1.000				
Q10	0.756	0.453	1.000			
Q13	0.386	0.583	0.780	1.000		
Q20	-0.247	-0.311	-0.039	-0.039	1.000	
Q23	0.047	0.016	0.286	0.462	0.393	1.000

Note: Data in bold indicate values with this range.

TABLE A13. Matrix of *P*-Values for Student Survey

	Leadership	Gender	Highest degree	Affiliated school/college/unit	Age Group
Q1	1.000	0.539	0.733	1.000	0.062
Q2	1.000	0.748	0.918	1.000	0.002
Q3	1.000	0.748	0.918	1.000	0.002
Q4	1.000	0.748	0.918	1.000	0.002
Q5	1.000	0.748	0.918	1.000	0.002
Q6	1.000	0.748	0.918	1.000	0.002
Q7	1.000	0.748	0.918	1.000	0.002
Q8	1.000	0.748	0.918	1.000	0.002
Q9	1.000	0.748	0.918	1.000	0.002
Q10	0.839	0.081	0.604	0.492	0.218
Q11	0.839	0.081	0.604	0.492	0.218
Q12	0.839	0.081	0.604	0.492	0.218
Q13	1.000	0.741	0.662	1.000	0.315
Q14	1.000	0.642	0.918	0.768	0.476
Q15	1.000	0.642	0.918	0.768	0.476
Q16	1.000	0.701	0.178	0.159	0.309
Q17	1.000	0.539	0.325	0.275	0.223
Q18	0.732	0.889	0.519	0.457	0.068
Q19	1.000	0.807	0.208	0.275	0.571
Q20	0.853	0.748	0.918	1.000	0.636
Q21	0.433	0.374	0.604	0.492	0.830
Q22	0.702	0.464	0.280	1.000	0.435
Q23	0.732	0.288	0.789	1.000	0.677

Note: Data in bold indicate *p*-values smaller than .5 significant value.

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